

Local and global well-posedness of the scalar Brazovskii gradient flow

TECT verification-first repository · claim A2-PDE-WELLPOSED

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1. Purpose and scope

This note establishes the well-posedness foundation A2 for the scalar Brazovskii gradient flow, on which the Sector-B free-energy minimisation rests dynamically. The argument is standard semilinear-parabolic theory (Henry 1981; Pazy 1983; Lunardi 1995). **Scope (deliberately narrow, operator review 2026-06-23):** the result proved here is

$$\text{A2}_{\text{scalar, periodic, } \mu^2 > 0, \gamma > 0} : \quad \frac{3}{2} < s \leq 2 \implies \text{global well-posedness}, \quad (1)$$

i.e. ONE real scalar field on a FIXED periodic cell with $\mu^2 > 0$, $\gamma > 0$. It does **not** cover the full Class-II multi-field action, nor the $\mu^2 < 0$ condensate branch of the numerical mainline; those are separate proof targets ($\text{A2}_{\text{full Class II}}$, $\text{A2}_{\mu^2 < 0}$). In particular $\mu^2 > 0$ is the *disordered-side* condition giving a positive linear gap; it is *not* a proof of BCC condensation. Numerical constants are pinned by [codes/foundations/a2_wellposedness_checks.py](#) (8/8 PASS).

2. Setup

Let $\mathbb{T}^3 = \mathbb{R}^3/\Lambda$ be the fixed periodic cell, Λ^* its reciprocal lattice. With the A1-KERNEL-CONV kernel $K(q) = \mu^2 + Y(q^2 - q_0^2)^2$ ($Y > 0$),

$$F_{\text{TECT}}[\phi] = \int_{\mathbb{T}^3} \left[\frac{1}{2} \phi K(-i\nabla) \phi + \frac{\lambda}{4} \phi^4 + \frac{\gamma}{6} \phi^6 \right] dx, \quad (2)$$

with L^2 gradient flow, on $X := L^2(\mathbb{T}^3)$,

$$\partial_t \phi = -L\phi + N(\phi), \quad L := K(-i\nabla), \quad N(\phi) := -\lambda \phi^3 - \gamma \phi^5, \quad \phi(0) = \phi_0. \quad (3)$$

Standing hypotheses **(H1)** $\mu^2 > 0$, **(H2)** $\gamma > 0$ (production anchor $\mu^2 = 5 \times 10^{-3}$, $\gamma = 1.62$).

3. The linear part generates an analytic semigroup

L is the self-adjoint Fourier multiplier $K(k)$, $k \in \Lambda^*$, with $D(L) = H^4$. The relevant spectral fact is the *lower bound*

$$\lambda_0 := \min_{k \in \Lambda^*} K(k) \geq \mu^2 > 0 \quad (\text{H1}), \quad (4)$$

since $K(k) = \mu^2 + Y(\dots)^2 \geq \mu^2$ for every k . Equality $\lambda_0 = \mu^2$ holds only if some lattice mode lies exactly on the shell $|k| = q_0$; on a generic cell $\lambda_0 > \mu^2$ (assert `lambda0_lower_bound_mu2_positive`: μ^2 is the sharp continuum infimum, lattice $\lambda_0 > \mu^2$). Only $\lambda_0 > 0$ is used: a positive self-adjoint operator is sectorial, so $-L$ generates an analytic semigroup e^{-tL} , fractional powers L^α exist with $D(L^\alpha) = H^{4\alpha} =: X^\alpha$, and

$$\|L^\beta e^{-tL}\|_{X \rightarrow X} \leq C_\beta t^{-\beta}, \quad C_\beta = \sup_{x \geq 0} x^\beta e^{-x} = \beta^\beta e^{-\beta} < 1 \quad (0 < \beta < 1). \quad (5)$$

Moreover $K(k) \asymp (1 + |k|^2)^2$ with constants depending on μ^2, Y, q_0 (lower constant positive by (4)), so $\langle \phi, L\phi \rangle$ is equivalent to $\|\phi\|_{H^2}^2$.

4. The nonlinearity is locally Lipschitz on X^β

Fix $\beta \in (\frac{3}{8}, \frac{1}{2}]$, so $X^\beta = H^{4\beta}$ with $\frac{3}{2} < 4\beta \leq 2$. By Sobolev in $d = 3$, $H^{4\beta} \hookrightarrow L^\infty$ for $4\beta > \frac{3}{2}$ (assert `sobolev_linf_threshold`), and $H^{4\beta}$ is a Banach algebra. Hence on balls $\{\|\cdot\|_{X^\beta} \leq R\}$,

$$\|N(\phi) - N(\psi)\|_X \leq \ell(R)\|\phi - \psi\|_{X^\beta}, \quad \ell(R) = C(|\lambda|R^2 + \gamma R^4), \quad (6)$$

so $N : X^\beta \rightarrow X$ is locally Lipschitz and bounded on bounded sets.

5. Local existence, uniqueness, regularity ($\frac{3}{2} < s \leq 2$)

With L sectorial (§3) and $N : X^\beta \rightarrow X$ locally Lipschitz (§4), Henry (1981, Thm. 3.3.3) gives, for every $\phi_0 \in X^\beta = H^{4\beta}$ ($s := 4\beta \in (\frac{3}{2}, 2]$), a unique mild solution

$$\phi \in C([0, \tau]; H^s) \cap C((0, \tau]; H^4) \cap C^1((0, \tau]; X), \quad \phi(t) = e^{-tL}\phi_0 + \int_0^t e^{-(t-\sigma)L}N(\phi(\sigma))d\sigma, \quad (7)$$

the integral converging by (5). Uniqueness and continuous dependence follow by Gronwall; parabolic smoothing plus smoothness of N bootstrap $\phi \in C^\infty((0, \tau) \times \mathbb{T}^3)$. **The range is $\frac{3}{2} < s \leq 2$:** the local theory used here employs only X^β , $\beta \leq \frac{1}{2}$. Persistence in H^s for $s > 2$ requires a separate higher-order estimate not carried out here and is therefore not claimed.

6. Global existence ($\frac{3}{2} < s \leq 2$)

F_{TECT} is a strict Lyapunov functional along the flow, $\frac{d}{dt}F[\phi(t)] = -\|\delta F/\delta\phi\|_{L^2}^2 \leq 0$, and under (H2) it is coercive: by Young $\frac{|\lambda|}{4}t^4 \leq \frac{\gamma}{12}t^6 + C_*$ ($C_* = 1.01 \times 10^{-2}$, §verification),

$$F_{\text{TECT}}[\phi] \geq \frac{1}{2}\langle \phi, L\phi \rangle + \frac{\gamma}{12}\|\phi\|_{L^6}^6 - C_*|\mathbb{T}^3|. \quad (8)$$

Energy domain (operator review #2). For $\frac{3}{2} < s < 2$ the quadratic form $\langle \phi_0, L\phi_0 \rangle$ need not be finite, so $F_{\text{TECT}}[\phi_0]$ may be undefined and $F[\phi(t)] \leq F[\phi_0]$ cannot be used at $t = 0$. Use parabolic smoothing. Let $T_{\max}$ be the maximal existence time and suppose $T_{\max} < \infty$; fix $t_0 \in (0, T_{\max})$. Smoothing (§5) gives $\phi(t_0) \in H^4(\mathbb{T}^3)$, hence $F_{\text{TECT}}[\phi(t_0)] < \infty$. On $[t_0, T_{\max})$ the energy identity gives $F[\phi(t)] \leq F[\phi(t_0)]$, and coercivity turns this into a uniform H^2 bound there ($\langle \phi, L\phi \rangle$ is equivalent to $\|\phi\|_{H^2}^2$, §3). Since $\beta \leq \frac{1}{2}$ ($4\beta \leq 2$) gives $H^2 \subseteq H^{4\beta} = X^\beta$, the X^β norm stays bounded on $[t_0, T_{\max})$, contradicting the continuation alternative. Hence $T_{\max} = \infty$ and the solution is global.

Theorem (A2, scalar/periodic/ $\mu^2 > 0/\gamma > 0$). Under (H1) and (H2), for every $\phi_0 \in H^s(\mathbb{T}^3)$ with $\frac{3}{2} < s \leq 2$ the flow (3) has a unique global solution $\phi \in C([0, \infty); H^s) \cap C^\infty((0, \infty) \times \mathbb{T}^3)$, depending continuously on ϕ_0 ; and $F_{\text{TECT}}[\phi(t)]$ is non-increasing for $t > 0$. If in addition $\phi_0 \in H^2$, then $F_{\text{TECT}}[\phi(0)] < \infty$ and the monotonicity holds for all $t \geq 0$.

7. Devil's-advocate (operator review 2026-06-23)

- (α) “ $\inf \text{spec } L = \mu^2$ is false on a generic cell.” UPHELD and fixed: the proof now uses only the lower bound $\lambda_0 := \min_k K(k) \geq \mu^2 > 0$ (4); equality is not claimed (it is non-generic). Sectoriality and the H^2 equivalence need only $\lambda_0 > 0$.
- (β) “The Sobolev inclusion is backwards.” UPHELD and fixed: for $\beta \leq \frac{1}{2}$ the correct direction is $H^2 \subseteq H^{4\beta} = X^\beta$ (§6), which still yields “ H^2 bound controls X^β ” — the conclusion survives in the correct direction.

- (γ) “The proof only closes for $\frac{3}{2} < s \leq 2$.” UPHELD: the theorem range is now stated as $\frac{3}{2} < s \leq 2$; $s > 2$ persistence is flagged as a separate higher-order estimate, not claimed. Scope narrowed to one scalar field on a fixed periodic cell, $\mu^2 > 0$ (disordered side, NOT BCC condensation), $\gamma > 0$; full Class-II and $\mu^2 < 0$ remain separate targets.
- (δ) “ $F[\phi_0]$ is undefined for $\frac{3}{2} < s < 2$, so $F[\phi(t)] \leq F[\phi_0]$ is illegitimate.” UPHELD and fixed (§6, operator review #2): the a priori bound is taken from $t_0 > 0$ via parabolic smoothing ($F[\phi(t)] \leq F[\phi(t_0)] < \infty$ on $[t_0, T_{\max})$), and energy monotonicity is stated for $t > 0$ (for $t \geq 0$ only under the extra assumption $\phi_0 \in H^2$).

Quantitative sanity check (mandatory): all constants reproduced by the 8/8 verification (v1.1): $\lambda_0 \geq \mu^2 = 5 \times 10^{-3}$ with strict $\lambda_0 > \mu^2$ on a generic lattice; Sobolev $s > 3/2$, $\beta > 3/8$; smoothing $C_\beta = \beta^\beta e^{-\beta} \in (0.43, 0.48)$; Young $C_* = 1.01 \times 10^{-2}$; continuation window $\beta \in (3/8, 1/2]$, $4\beta \leq 2$, $H^2 \subseteq X^\beta$.

8. Result footer

Result ID: A2-PDE-WELLPOSED (scalar, periodic, $\mu^2 > 0$, $\gamma > 0$)

Precise statement: Under $\mu^2 > 0$ and $\gamma > 0$, the scalar Brazovskii L2 gradient flow on a fixed T^3 is globally well-posed for ϕ_0 in H^s , $3/2 < s \leq 2$: unique global solution, smooth for $t > 0$, continuous dependence, F non-increasing for $t > 0$.

Scope: ONE real scalar field, fixed periodic cell, $3/2 < s \leq 2$. $\mu^2 > 0$ = disordered-side positive linear gap (NOT BCC condensation). NOT full Class-II; NOT $\mu^2 < 0$ branch.

Dependencies: A1-KERNEL-CONV.

Hypotheses: (H1) $\mu^2 > 0$ [$\lambda_0 = \min_k K(k) > \mu^2 > 0$: sectoriality + H^2 equivalence]; (H2) $\gamma > 0$ [coercivity, global].

Evidence grade: PROVED CONDITIONAL (analytic) + EXECUTED (8/8 checks).

Reproduction command: python codes/foundations/a2_wellposedness_checks.py

Expected output: 8/8 PASS; exit 0.

Falsification gate: finite-time blow-up or non-uniqueness for some ϕ_0 in H^s ($3/2 < s \leq 2$) at $\mu^2 > 0$, $\gamma > 0$; or $\lambda_0 = \min_k K(k) \leq 0$.

Tier before / after: A2 T1 / T6 PROVED CONDITIONAL (scalar/periodic/ $\mu^2 > 0$ scope) -- RE-SUBMITTED for operator sign-off after the 2026-06-23 review #1+#2.

No-overclaim statement: No $s > 2$ persistence; no full Class-II; no $\mu^2 < 0$ condensate branch; $\mu^2 > 0$ is NOT BCC condensation; no unique global minimiser (that is Sector B).

Next required action: Operator sign-off of the narrowed T6; then A3 (renormalisation / continuum limit); separately A2_full-ClassII and A2_ $\{\mu^2 < 0\}$.